SE2020 – Web and Mobile Technologies

Faculty of Computing

SLIIT

Lab Test

2 Hours

Item Manager MERN Application Enhancement and Deployment

1. Lab Test Overview

You are provided with a starter MERN Item Manager project containing separate frontend and backend applications. Your task is to complete the given project, extend it by adding a new field called Discount Percentage, test it locally using your own MongoDB connection string, publish the source code to a public GitHub repository, and deploy the frontend and backend separately using free hosting platforms.

2. Expected Skills Demonstrated

- Understanding and completing an existing MERN project structure.
- Updating both frontend and backend to support a new data field.
- Testing the system locally using MongoDB Atlas or an equivalent MongoDB connection.
- Using Git and GitHub to publish the full project in a public repository.
- Deploying frontend and backend separately and connecting the live frontend to the live backend.

3. Resources Provided

- Frontend project of the Item Manager application.
- Backend project of the Item Manager application.
- Basic CRUD functionality already included in the starter project.
- You are expected to complete, test, and deploy the provided codebase during the lab test.

4. Tasks to Be Completed

Task 01 – Project setup and local verification

• Extract the provided **Item Manager project** (Frontend & Backend).

Backend Setup

• Open the backend project in a terminal or VS Code.
• Install required dependencies:

```
npm install
```

• Configure the environment variables:

- Add your **MongoDB Connection URL** in the .env file
- Example:

```
MONGO_URI=your_mongodb_connection_string
PORT=5000
```

• Start the backend server:

```
npm run dev
```

• Ensure the backend runs without errors.

Frontend Setup

• Open the frontend project in a new terminal or VS Code.
• Install dependencies:

```
npm install
```

• Start the frontend application:

```
npm run dev
```

Verification

• Open the application in your browser (http://localhost:5173).
• Verify the following:

- Application loads successfully
- You can navigate between pages
- Existing features (Add Item / View Items) are working
- Capture screenshots of the local running existing system.

Task 02 – Functional enhancement

- Modify the application by adding a new field named Discount Percentage to the Add Item form.
- Update the relevant frontend component(s) to collect this value from the user.
- Update the backend model, controller logic, and API flow as needed so the value is saved correctly.
- Ensure that the Discount Percentage value is shown on the Home page together with the other item details.
- Make sure the new field behaves correctly during add, read, and any related update operations.
- Capture screenshots of the local running updated system as instructed in the submission section.

Task 03 – GitHub publication

- Create a new **public GitHub repository** using your own GitHub account *(Use your SLIIT GitHub account. If there are any issues, you may use your personal GitHub account.)*

- You may choose **ONE of the following approaches**:

  - **Option 01:** Use a **single repository** containing both frontend and backend
  - **Option 02:** Use **two separate repositories** (one for frontend and one for backend)

- Push the complete project to GitHub **after testing locally**
- Ensure:

  - The repository structure is clean and well organized
  - Both frontend and backend codes are clearly separated (if using a single repository)

- The GitHub repository/repositories **must be created before starting deployment**

Task 04 – Separate deployment

- Deploy the backend to a free hosting platform such as Railway, Render or another equivalent service.
- Deploy the frontend to a free hosting platform such as Netlify, Vercel or another equivalent service.
- Update the frontend configuration so that it communicates with the deployed backend URL.
- Verify that the live system works successfully after deployment.

5. Required Contents of the Submission Word Document

Your submitted Word document must include all of the following items clearly and in order:

- Student Name
- Student ID
- GitHub Repository Links
- Screenshot of the locally running existing project Add Item page
- Screenshot of the locally running existing project Home page
- Screenshot of the locally running updated project Add Item page
- Screenshot of the locally running updated project Home page
- Frontend Live URL
- Backend Live URL
- Brief deployment notes including the platforms used

6. Marking Scheme (100 Marks)

Component	Description	Marks	Comments for Students
Local setup and MongoDB configuration	Correctly configures the connection URL and verifies the project locally before deployment.	10	System should run without major local errors.
New field integration	Adds the new field to the form and updates the relevant frontend and backend logic correctly.	20	This is the main development task in the lab test.
Home page display and application behavior	New field is shown properly on the Home	10	Visible evidence must match the

Component	Description	Marks	Comments for Students
	page and the overall flow works correctly.		implementation.
GitHub repository management	Creates a public repository and pushes the complete project in a clear structure.	10	Repository must be accessible to the examiner.
Backend deployment	Deploys the backend successfully and provides a valid live backend URL.	20	Backend should support the required live requests.
Frontend deployment and integration	Deploys the frontend successfully and connects it to the live backend correctly.	25	Live application should function end to end.
Submission document quality	Includes all requested details, screenshots, and deployment notes clearly.	5	Missing evidence will reduce marks.

7. Important Rules

- This is an individual lab test.
- Use only your own GitHub account and your own deployment accounts.
- Do not remove existing core functionality from the provided starter project.
- Do not submit broken, inaccessible, or private links.
- Make sure screenshots are readable inside the submission document.